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Circuits Lab Experiment 3

Applications of Diode

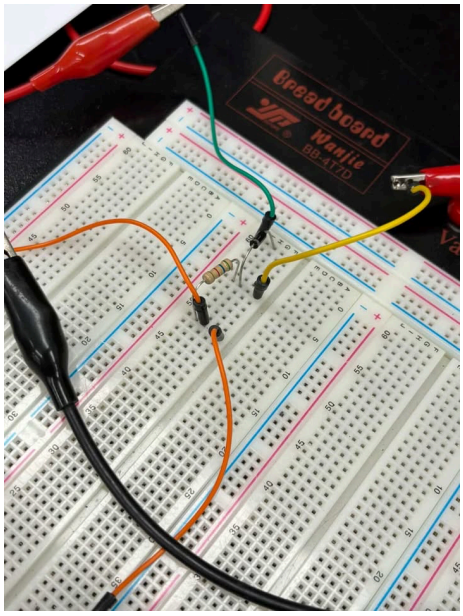
Objective:

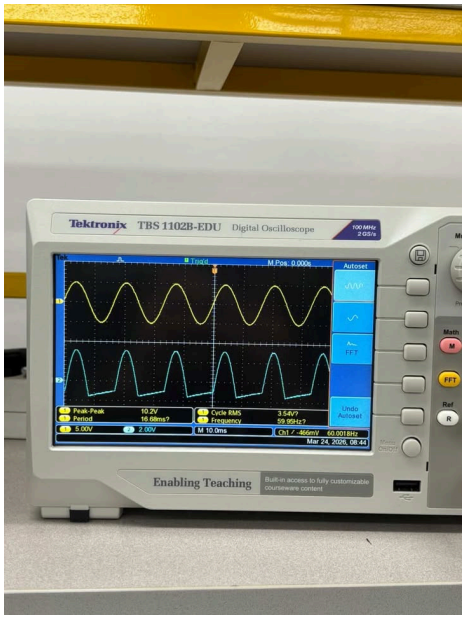
Study and observe the characteristics of diodes in electrical circuits, and see their applications. We investigate the properties of diodes, specifically the forward voltage drop. We learn to use diodes to regulate the voltage, protect circuits from a surge, and converting AC current to DC current.

Materials:

Component	Quantity
2.7 Ωk resistor	1
5.6 Ωk resistor	1
10 μF capacitor	1
22 μF capacitor	1
1N4004 diode	4

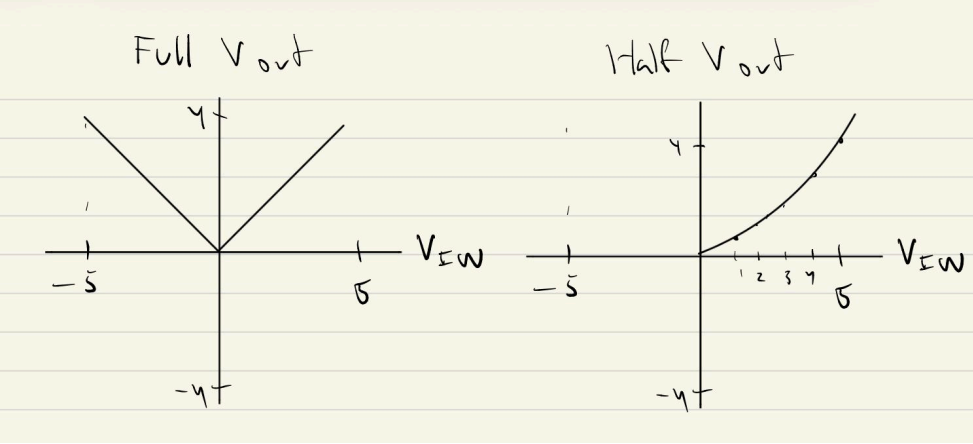
Half wave rectifier Experiment:



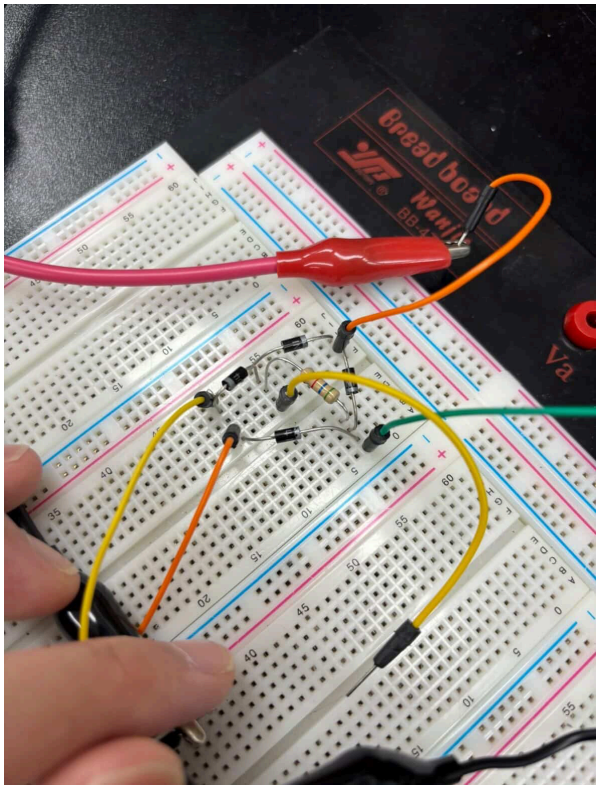


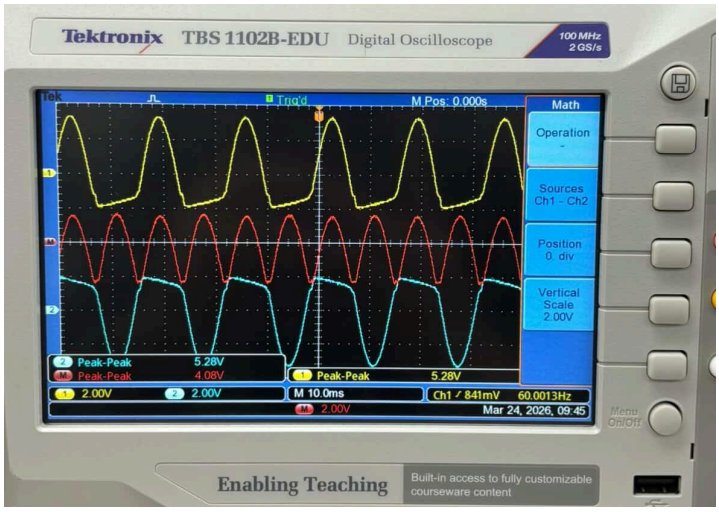
Vin [V]	Half Vout	Full Vout
-5	0	3.867
-4	0	2.895
-3	0	1.935
-2	0	0.997
-1	0	0.153
0	0	0
1	0.489	0.123
2	1.436	0.960
3	`2.406	1.899
4	3.379	2.860
5	4.345	3.831

Vin-Vout graph with Vin from -5V to 5V
Vdon = 0.5



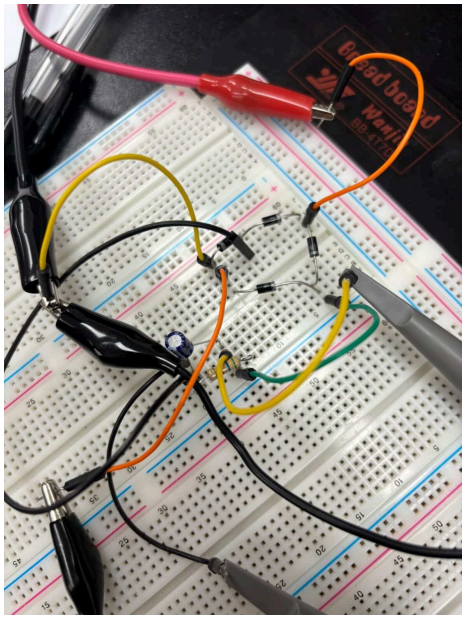
Rectifier Experiment

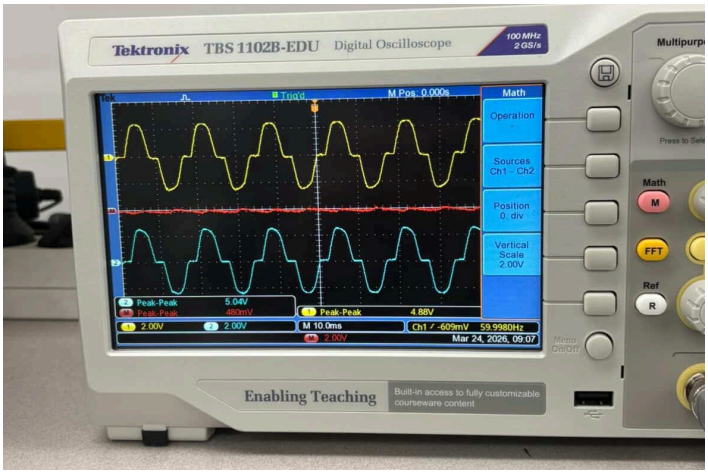




Question 8: The pros of using a full wave rectifier allow you to use both ends of the AC signal instead of losing it as seen when using a half wave rectifier. Full wave rectifiers are the solution.

Ripple in full wave rectifier Experiment:

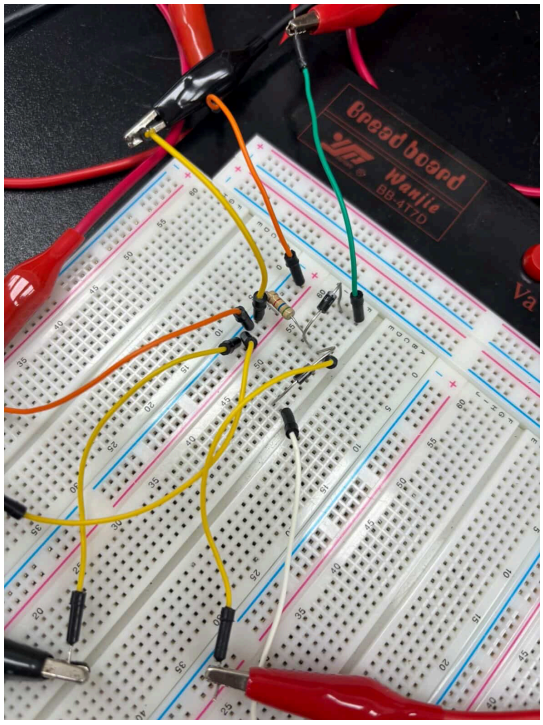


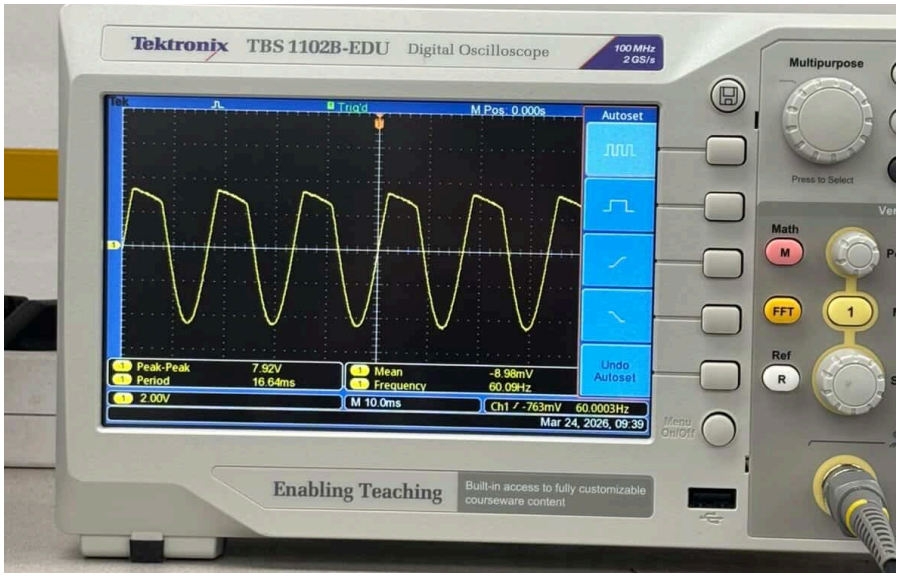


	Calculated Vr [mV]	Measured Vr [mV]
RL=5.6k.2,C,=22uRL=5.6k.2,	606.42	480
RL=2.7k2,C,=22uF, fin=60Hz	1256. 63	560
R_=5.6k2,C,=10uF, fin=60Hz	1322.45	640
RL=5.6k2,C,=22uF,fin=120Hz	309.26	320

It is possible that there were some issues with our measuring devices as the measured to calculated are quite different. It is possible that some of our variables could be off in either experimental or ideal.

Limiter Experiment:

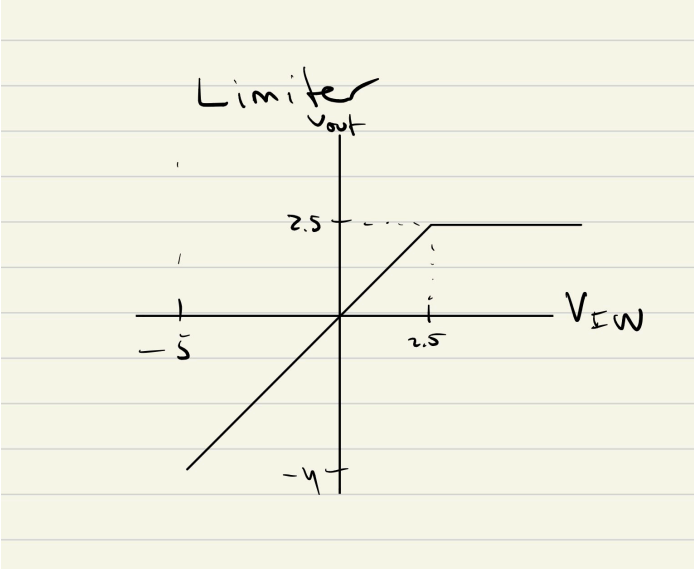




Vin [V]	Vout [V]
-5	-4.987
-4	-3.988

-3	-2.990
-2	-1.991
-1	-0.992
0	0
1	0.992
2	1.991
3	2.472
4	2.520
5	2.547

Graph of Vin/Vout Limiter

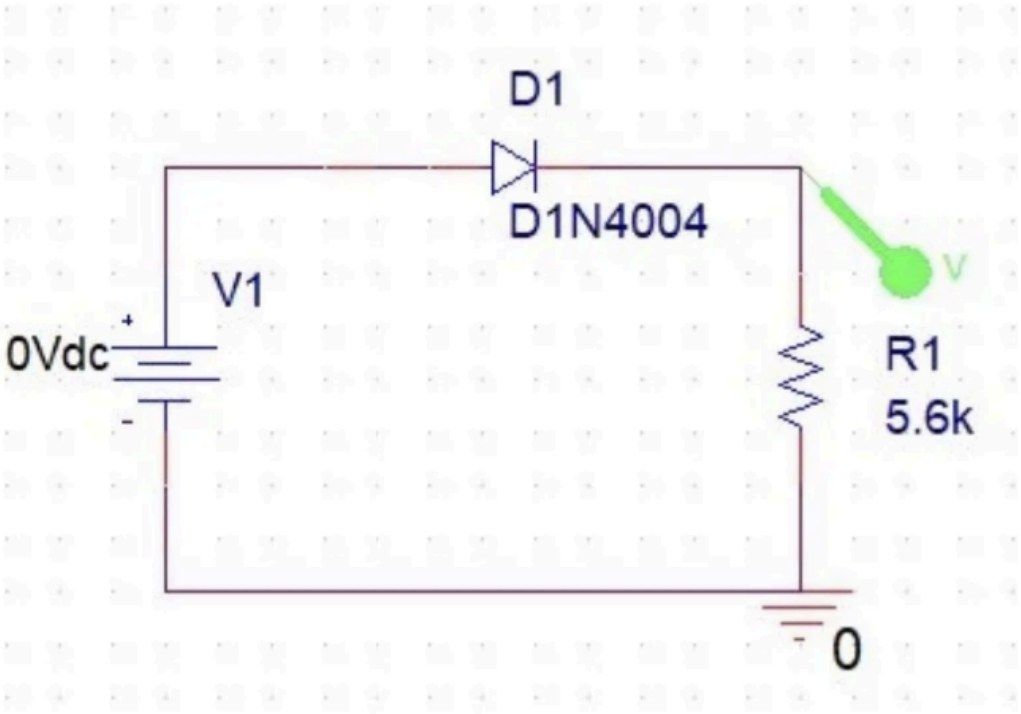


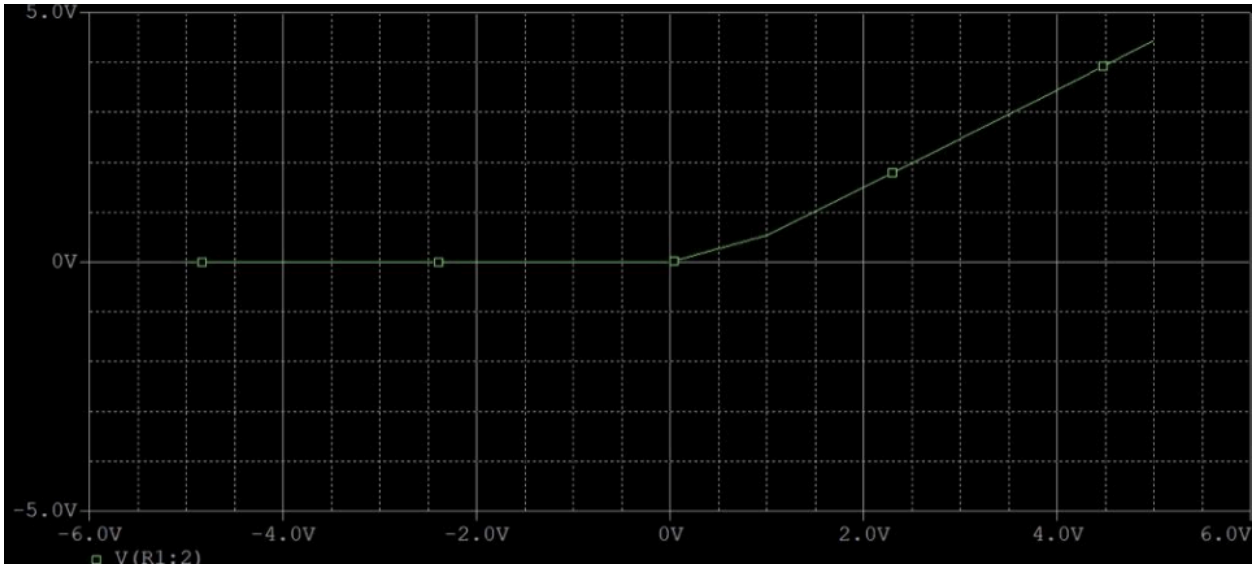
Vdon1: unknown
Vdon2: 2.5 V

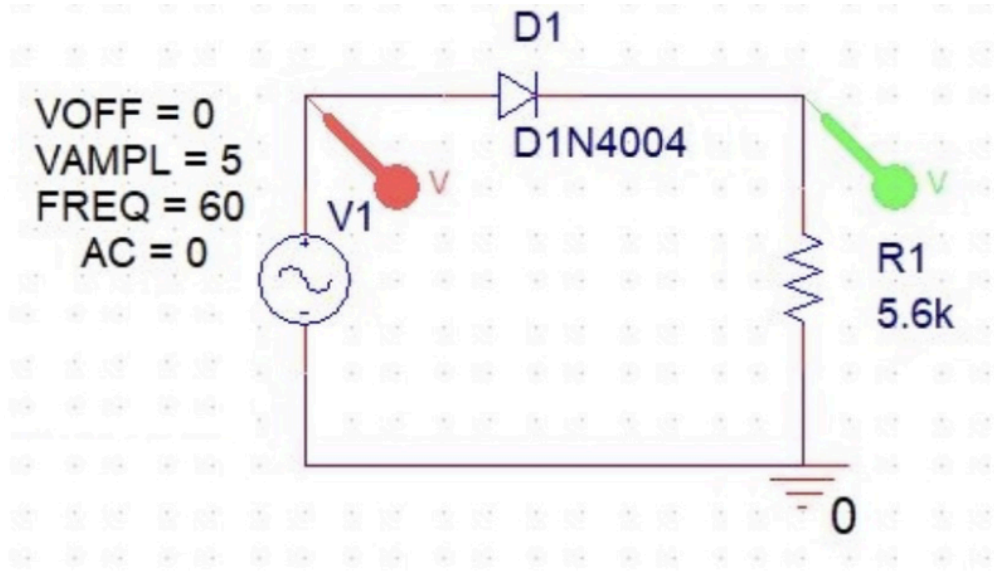
In theory, the limiter graph should be symmetrical over the origin, however, there seems not to be a limiter in the negative values. Perhaps the connections on the multimeter or in the breadboard were off.

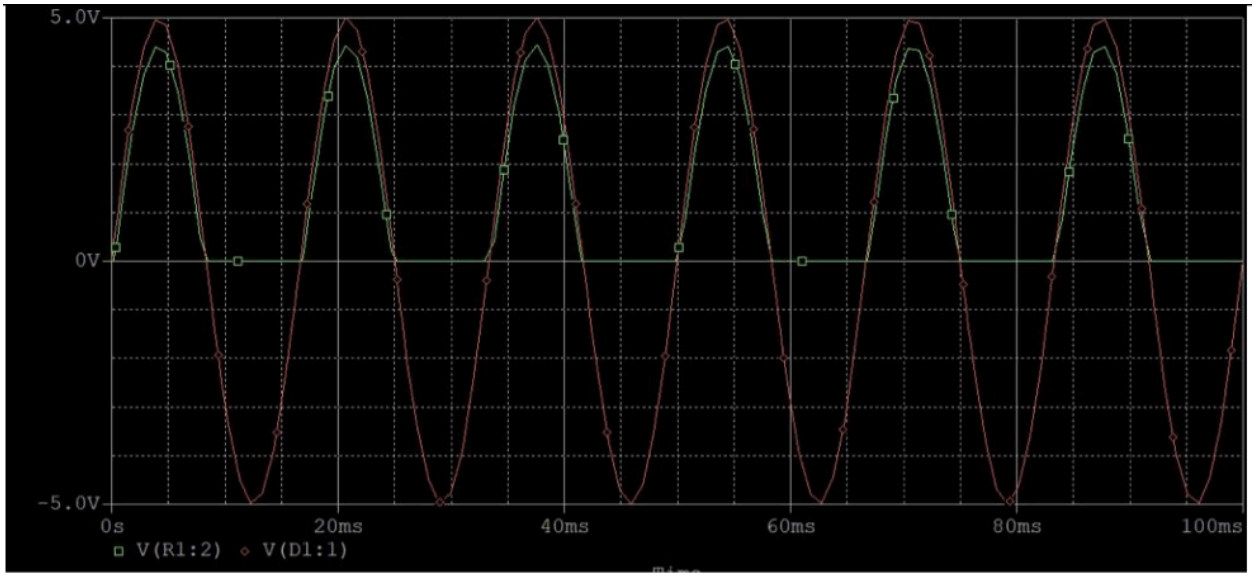
Pspice:

Experiment 1: Half wave rectifier

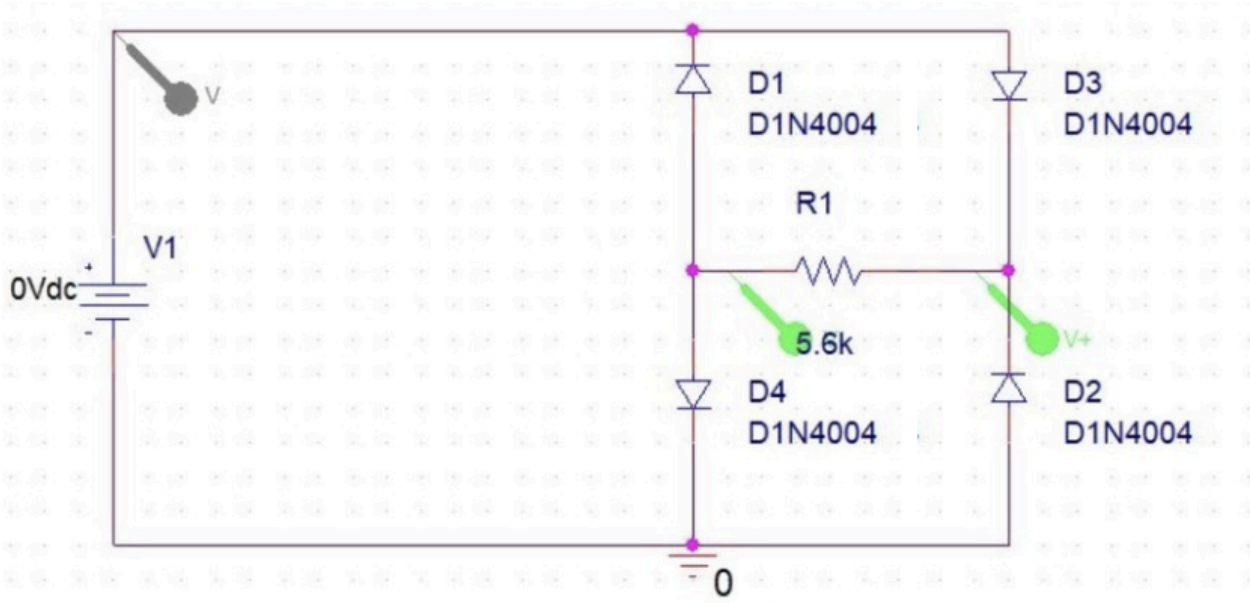


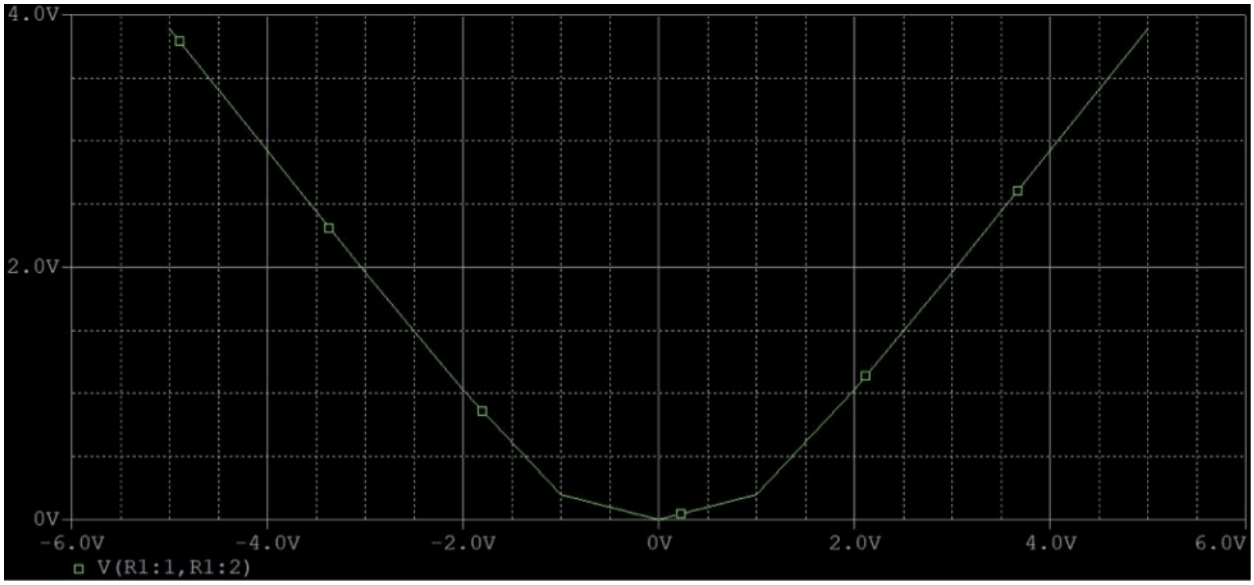

5.0V
0V
-5.0V
-6.0V
-4.0V
-2.0V
0V
2.0V
4.0V
6.0V
V(R1:2)

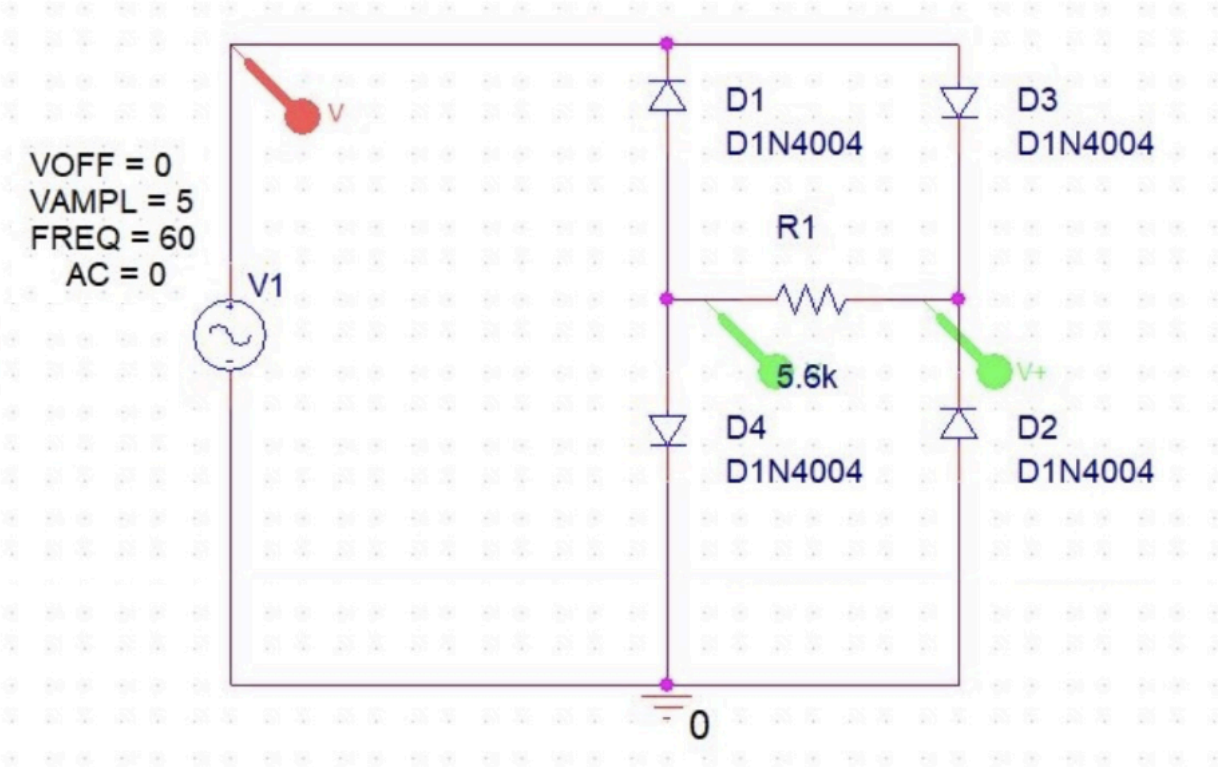

D1
D1N4004
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
V1
V
V
R1
5.6k
0

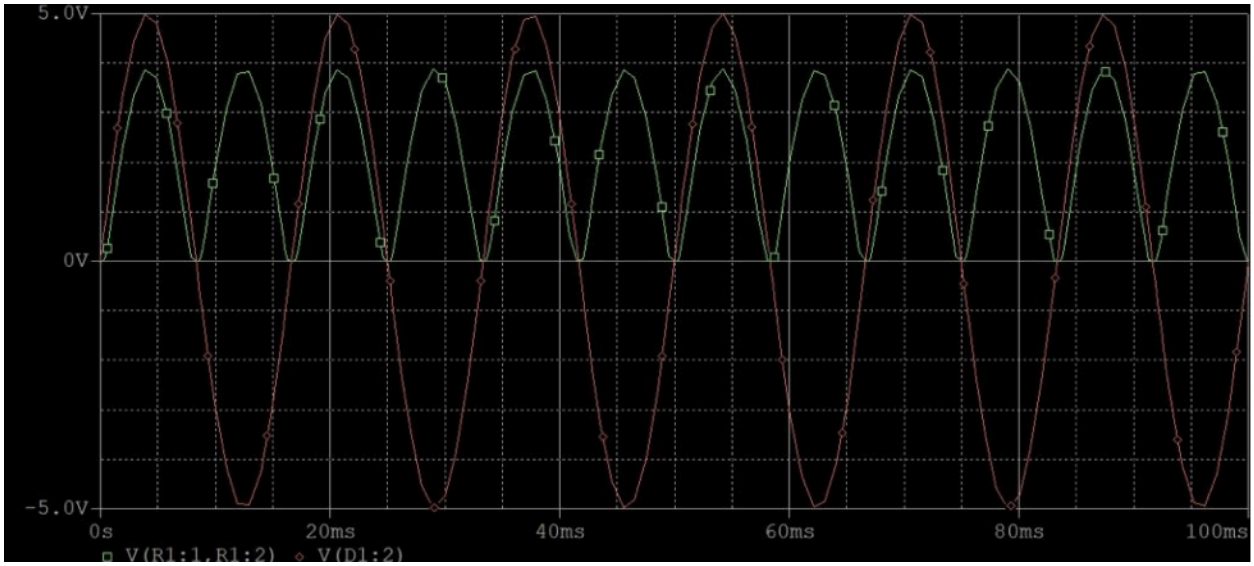


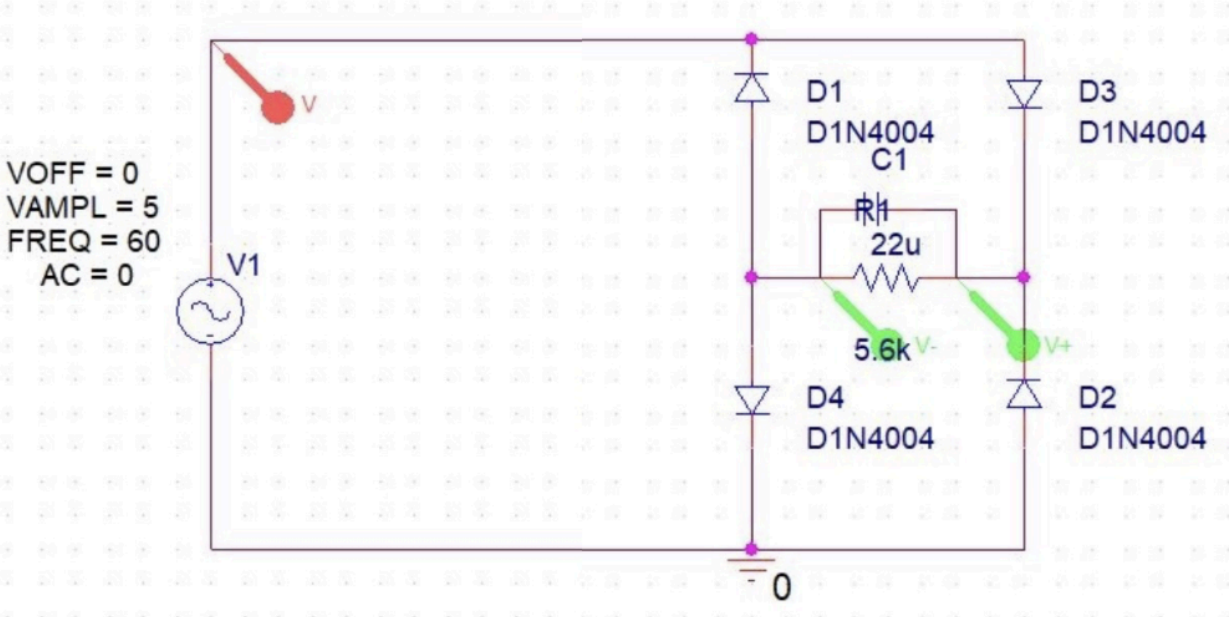
Experiment 2: Rectifier

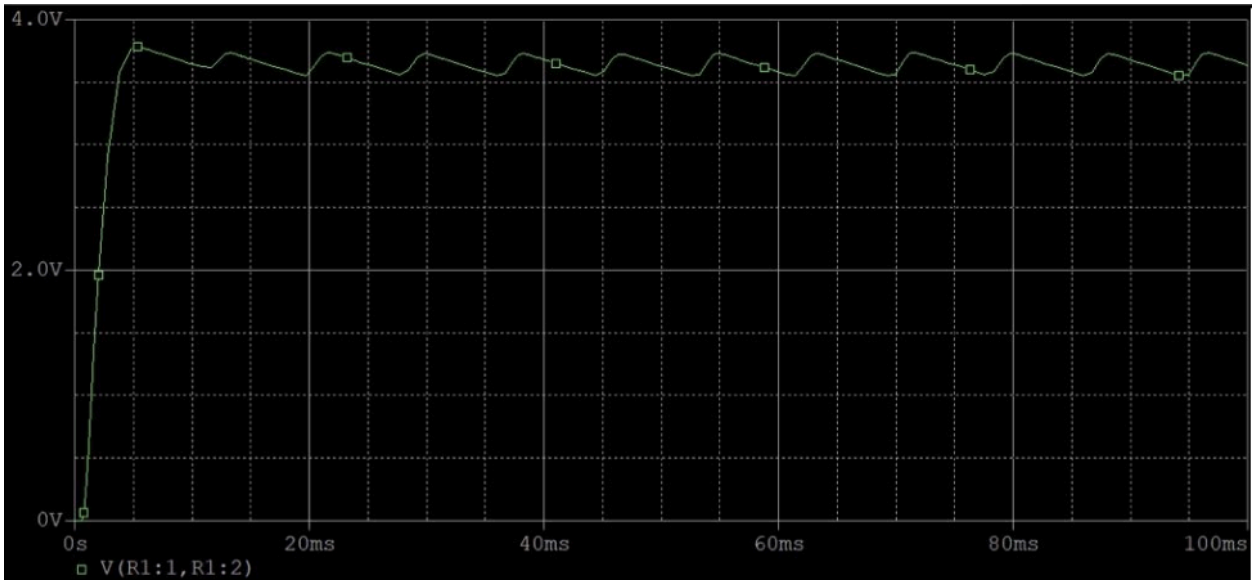


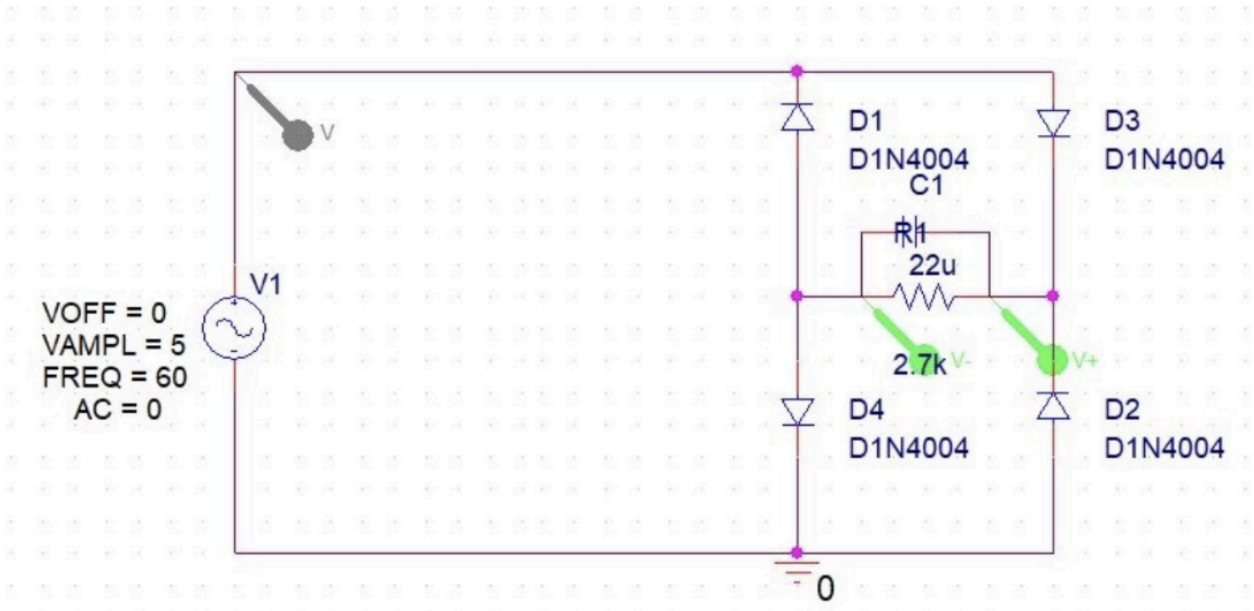

4.0V
2.0V
0V
-6.0V
-4.0V
-2.0V
0V
2.0V
4.0V
6.0V
V(R1:1,R1:2)

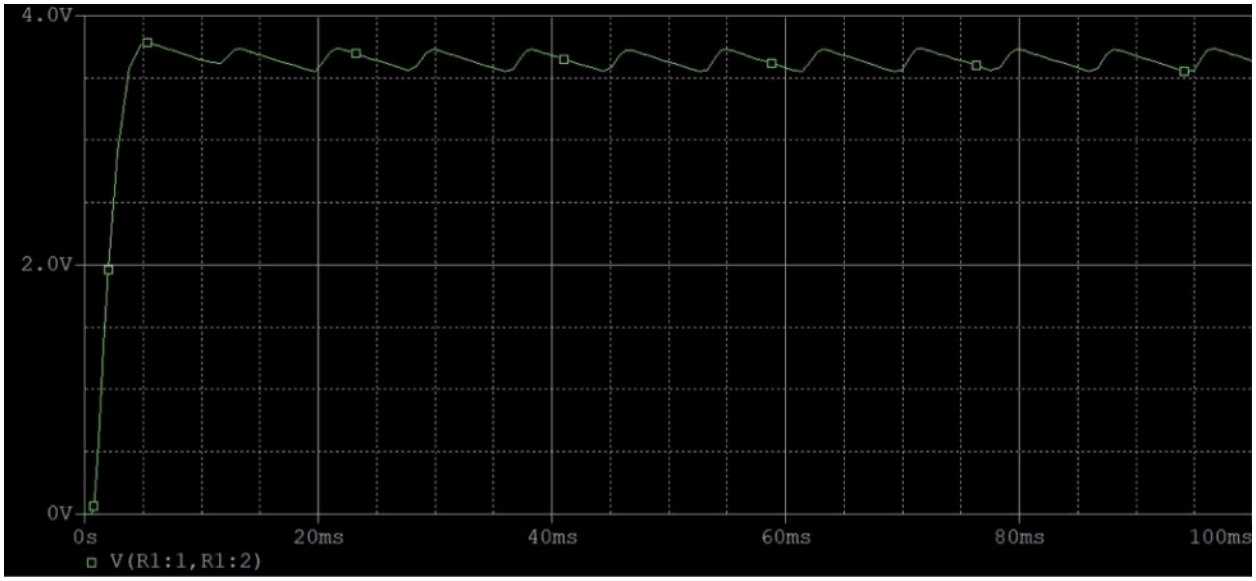

V
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
V1
D1
D1N4004
D3
D1N4004
R1
5.6k
V+
D4
D1N4004
D2
D1N4004
0

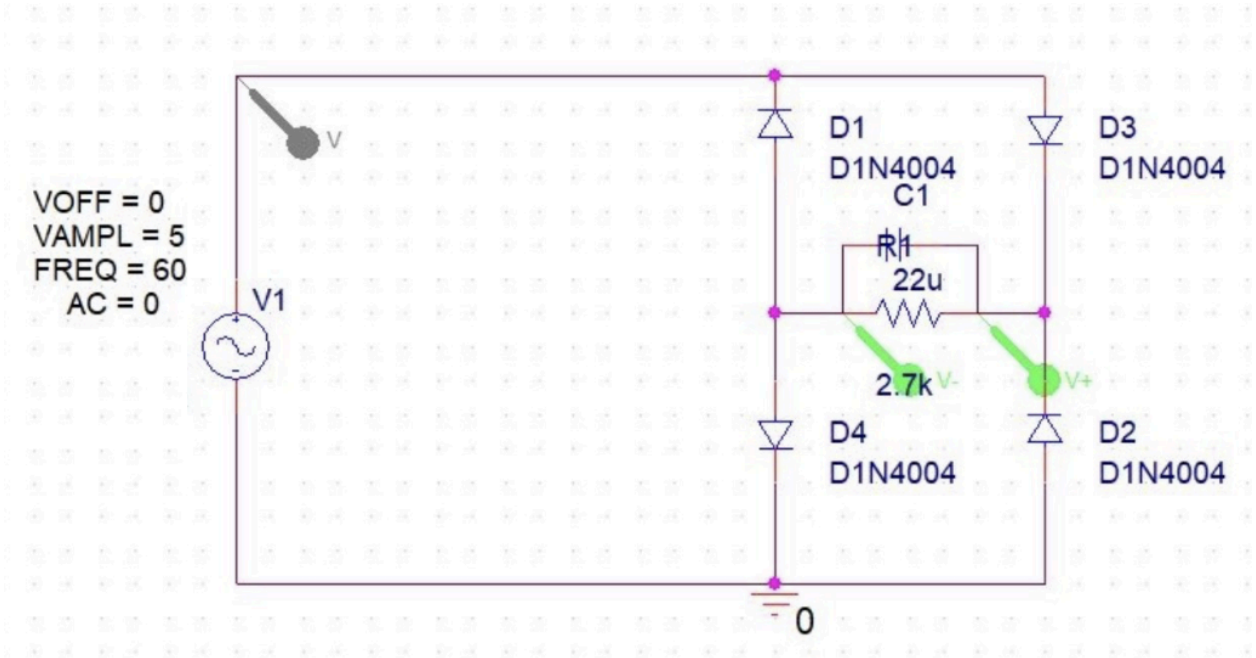

5.0V
0V
-5.0V
0s
20ms
40ms
60ms
80ms
100ms
V(R1:1,R1:2)
V(D1:2)

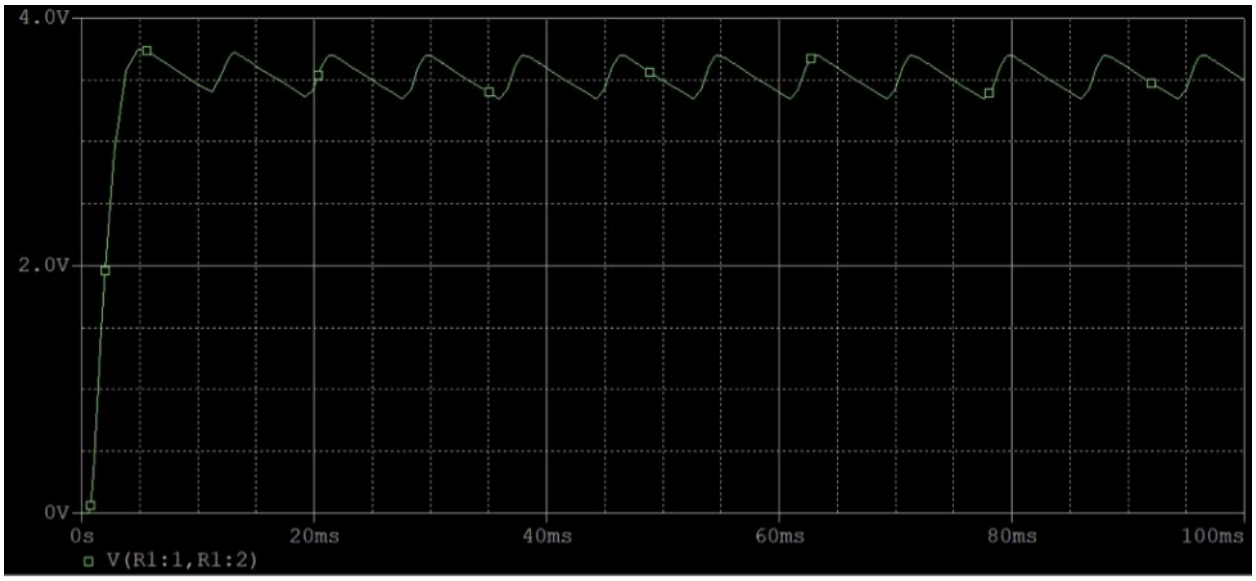

V
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
V1
D1
D1N4004
C1
R1
22u
5.6k
V-
V+
D3
D1N4004
D4
D1N4004
D2
D1N4004
0

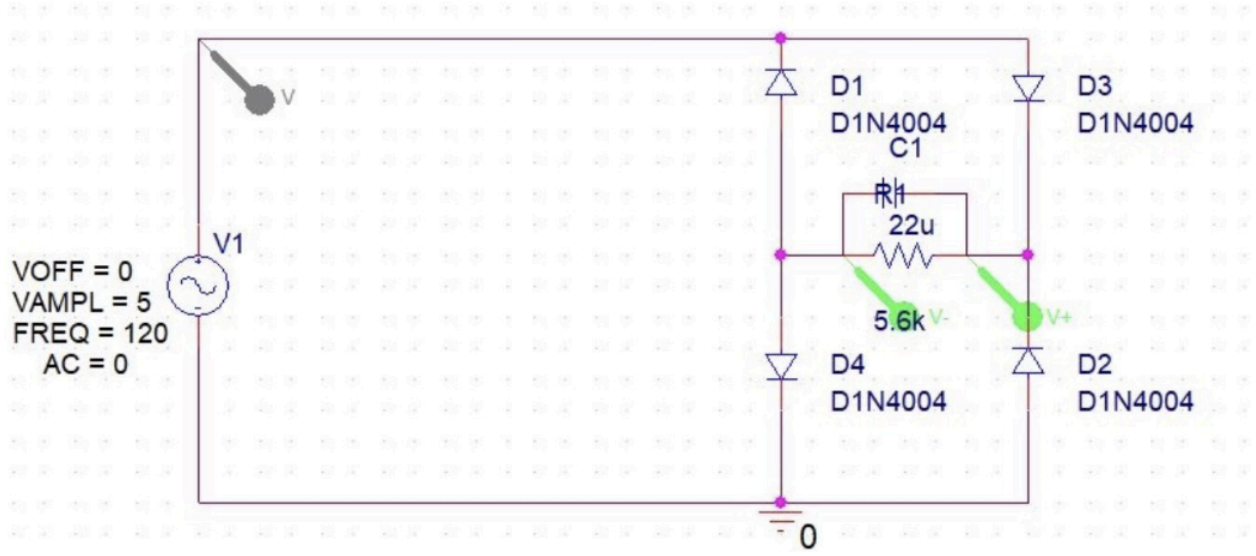

4.0V
2.0V
0V
0s
20ms
40ms
60ms
80ms
100ms
V(R1:1,R1:2)

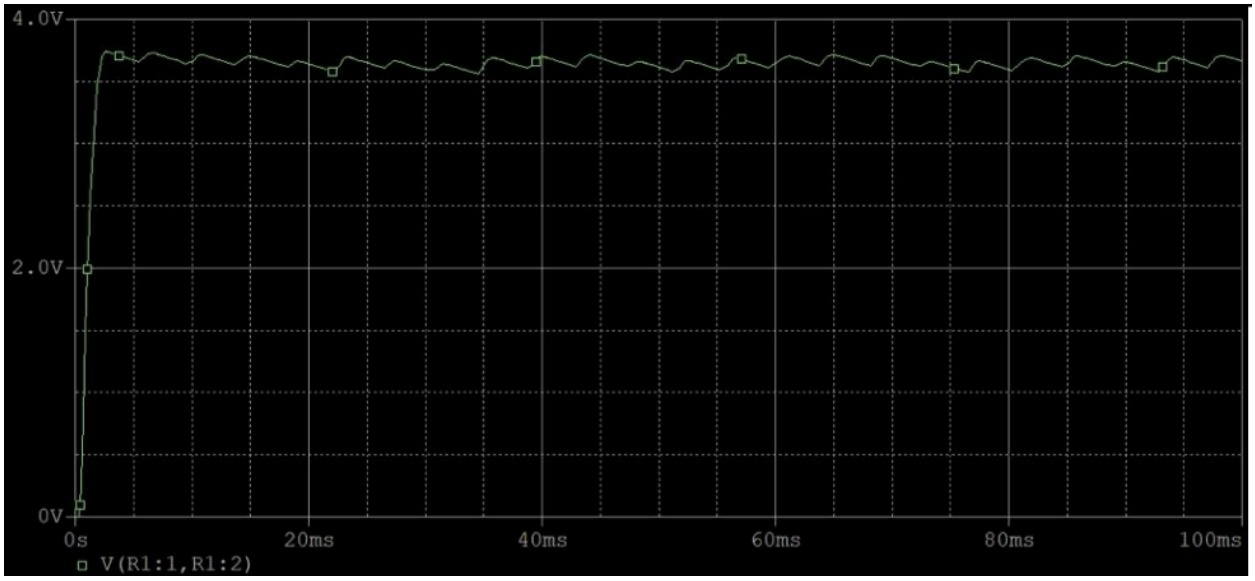

V
V1
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
D1
D1N4004
C1
22u
2.7k
V-
V+
D3
D1N4004
D4
D1N4004
D2
D1N4004
0


4.0V
2.0V
0V
0s
20ms
40ms
60ms
80ms
100ms
V(R1:1,R1:2)

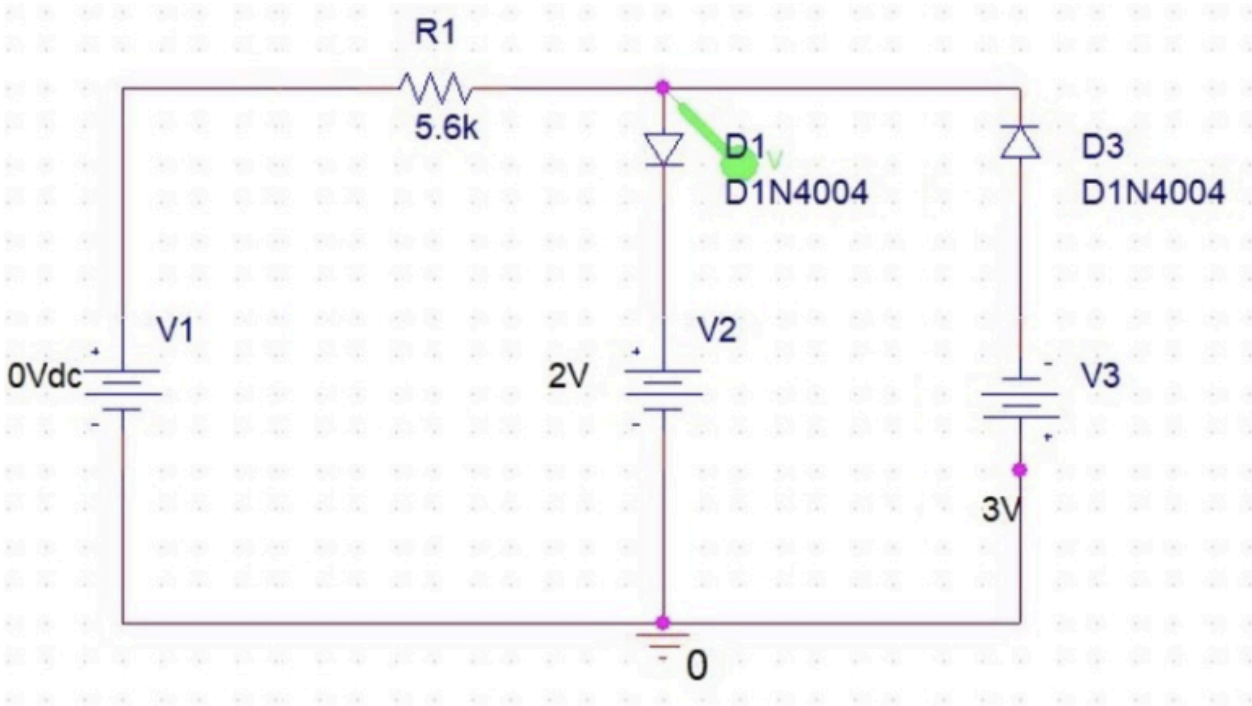

V
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
V1
D1
D1N4004
C1
R1
22u
2.7k
V-
V+
D3
D1N4004
D4
D1N4004
D2
D1N4004
0

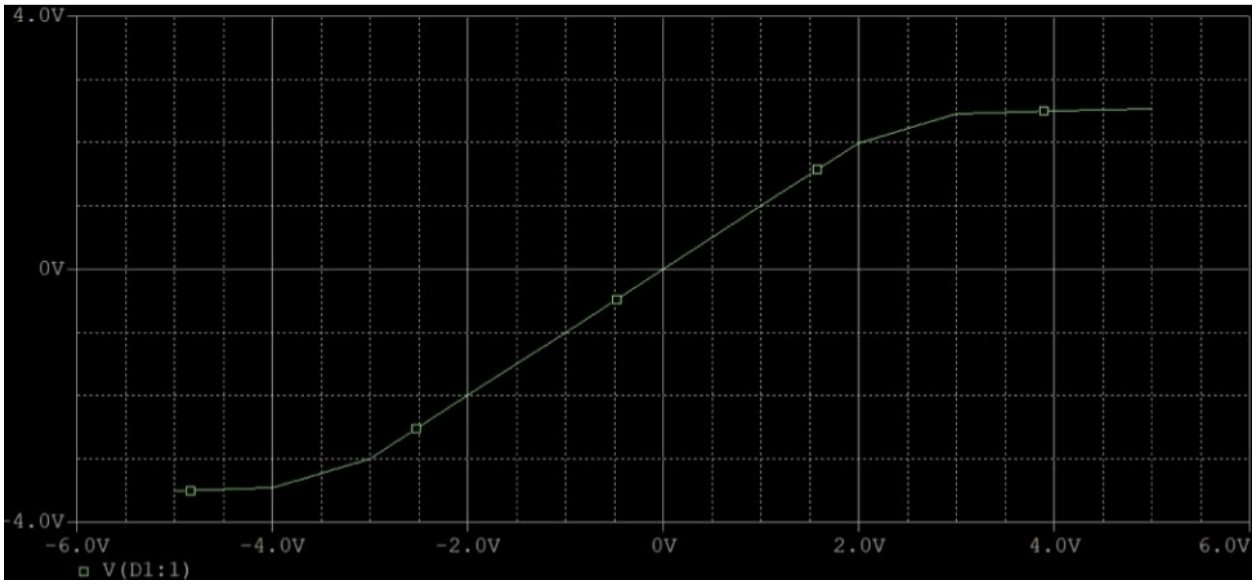

4.0V
2.0V
0V
0s
20ms
40ms
60ms
80ms
100ms
V(R1:1,R1:2)

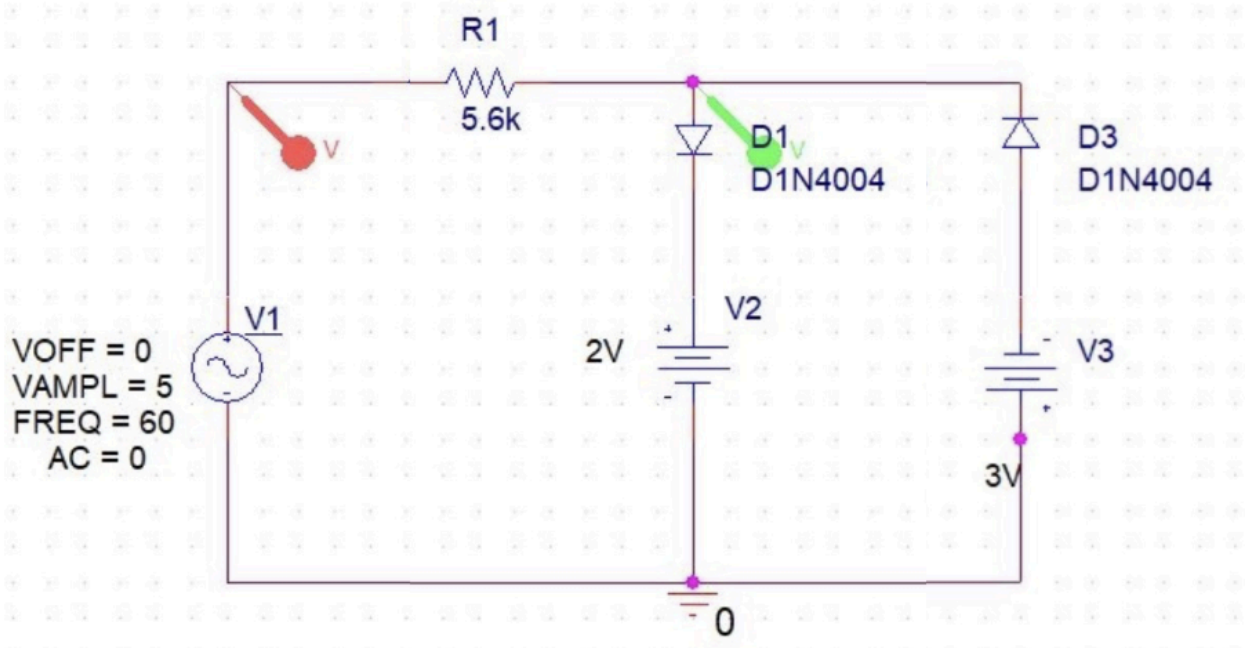

V
V1
VOFF = 0
VAMPL = 5
FREQ = 120
AC = 0
D1
D1N4004
D3
D1N4004
C1
22u
5.6k
V-
V+
D4
D1N4004
D2
D1N4004
0

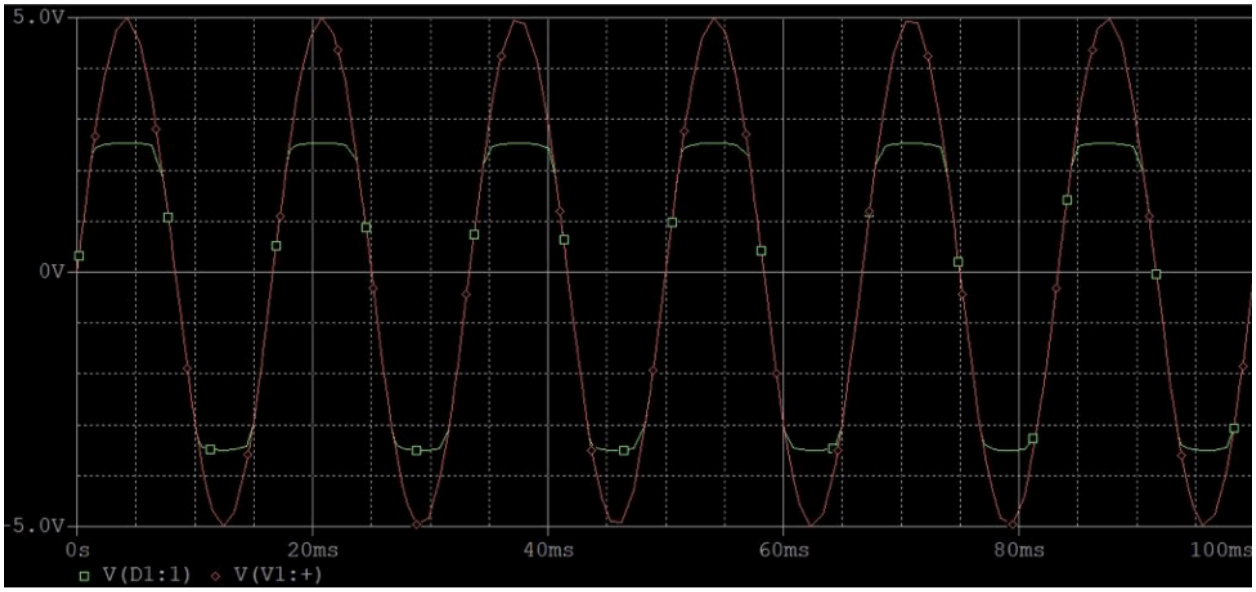


Experiment 3: Limiter Experiment




4.0V
0V
-4.0V
-6.0V
-4.0V
-2.0V
0V
2.0V
4.0V
6.0V
V(D1:1)


R1
5.6k
V
D1
D1N4004
D3
D1N4004
V1
VOFF = 0
VAMPL = 5
FREQ = 60
AC = 0
V2
2V
V3
3V
0


5.0V
0V
-5.0V
0s
20ms
40ms
60ms
80ms
100ms
V(D1:1)
V(V1:+)